

CC-Link Variable Speed Drive (VSD) Motor MAPS Template (Electrical)

CC-Link Variable Speed Drive (VSD) Motor MAPS template is designed to control a VSD motor that is connected via a CC-Link network.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the CC-Link VSD motor.

This document describes the Advanced CC-Link VSD Motor MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the CC-Link VSD motor being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

This document also provides a troubleshooting section describing the error descriptions associated with the various error codes and their error numbers.

Advanced CC-Link VSD Motor Functional Philosophy

The VSD start is enabled once all interlocks are ready and a start request is received.

Field Interlocks

- MCC (Overload) Trip Feedback
- Isolator On Feedback
- Safety Interlock Feedback
- VSD Ready Feedback
- Motor Current Feedback (Current above High Limit Setpoint)

PLC Interlocks

- Start Interlock Ready
- Process Interlock Ready
- Restart Delay Interlock
- There are no interlock faults latched

Motor Current – the motor current is monitored constantly after the running setpoint time has elapsed, if the current goes above the high current limit for five seconds the motor will trip on overcurrent.

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the VSD to become ready for operation.

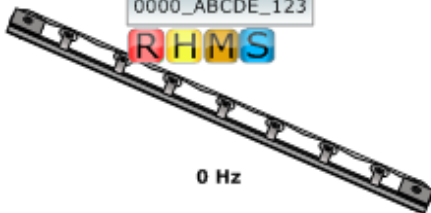
Maintenance mode– will disable the Start and Process Interlocks, but respects all the other interlocks. This is used to test a VSD while the process is stopped.

Simulation mode– will disable the physical output of the VSD and simulate an input run feedback after the elapsed setpoint time from the faceplate. The Start and Process Interlocks are respected. Simulation mode is used to test the process logic without starting the physical device itself. It is very useful in testing, commissioning and maintenance.

Start and stop requests are initiated by selecting a mode on the VSD faceplate:

- **Automode**- the auto control program initiates a start and stop, normal PLC control.
- **Manual mode& Desk mode**- the operator can start and stop from the faceplate.
- **Manual mode& Field mode**- the motor can be started and stopped from the start and stop push buttons in the field.

Typical Faceplate Graphics for the Advanced CC-Link VSD Motor MAPS Template

VSD ISO	VSD Motor Left	VSD Pump Left	VSD Conveyor Angle Left
0000_ABCDE_123 IRHSM M Frequency	0000_ABCDE_123 RHMS  0 Hz	0000_ABCDE_123 RHMS  0 Hz	0000_ABCDE_123 RHMS  0 Hz
H	Manual Mode		
M	Maintenance Mode		
S	Simulation Mode		
F	Running Forward		
R	Running Reversed		
0 Hz	Current Motor Frequency		
I	Process Interlock		
0000_ABCDE_123	Motor Control Name		
Graphics legend:			
Line colour	Fill colour	VSD status	
Black	Grey	Stopped	
Grey	White	Stopping	
Green	White	Starting	
Black	Green	Running	
Black	Yellow/Red	Fault Active	

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	Fail To Start Setpoint	Analog	x	x	x
SSCL 2	Fail To Stop Setpoint	Analog	x	x	
SSCL 3	Running Setpoint	Analog	x	x	x
SSCL 4	Stopping Setpoint	Analog	x	x	
SSCL 5	Simulate Setpoint	Analog	x	x	x
SSCL 6	Restart Setpoint	Analog	x	x	
SSCL 7	Max Current Setpoint (High Limit)	Analog	x	x	x
SSCL 8	Frequency Setpoint	Analog	x	x	x
SSCL 9	Parameter Read/Write Number	Analog	x		

SSCL 10	Parameter Read/Write Value	Analog	x		
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Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x
SSSL 1	Starting Time Process Value (actual)	Analog	x	x	x
SSSL 2	Stopping Time Process Value (actual)	Analog	x	x	
SSSL 3	No of Operations (actual)	Analog	x	x	
SSSL 4	Restart Delay Time Remaining (actual)	Analog	x	x	
SSSL 5	Motor Current Process Value (actual)	Analog	x	x	x
SSSL 6	Total Running Hours	Analog	x		
SSSL 7	VSD Status Word	Marshal	x	x	x
SSSL 8	Output Frequency	Analog	x	x	x
SSSL 9	VSD Alarm Code	Analog	x		

Digital Inputs:		Advanced	Standard	Basic
DI 0	Running Signal (On = Healthy)	x		
DI 1	Isolator Healthy (On = Healthy)	x	x	x
DI 2	MCC Healthy (On = Healthy)	x	x	x
DI 3	Safety Interlock Healthy (On = Healthy)	x	x	
DI 4	Manual Forward Start Push Button (On = Start)	x	x	
DI 5	Manual Reverse Start Push Button (On = Start)	x	x	
DI 6	Manual Stop Push Button (Off = Stop)	x		
Intelligent Module Control/Status Words (PPO2)				
10	Control Words	x	x	x
10	Status Words	x	x	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Auto/Manual	x	x	x	x
Bit 1	Field/Desk	x	x	x	x
Bit 2	Maintenance	x	x	x	x
Bit 3	Simulation	x	x	x	x
Bit 4	Alarms All Reset		x	x	x
Bit 5	Reset No Operations Accumulator	x	x	x	
Bit 6	Start Command	x	x	x	x
Bit 7	Reset Running Hours Accumulator	x	x		
Bit 8	Stop Command	x	x	x	x
Bit 9	External Control	x	x	x	x
Bit 10	Start Forward	x	x	x	x
Bit 11	Start Reverse	x	x	x	x
Bit 12	Read Parameter		x		
Bit 13	Write Parameter		x		
Bit 14	VSD Alarm Reset		x		
Bit 15	Spare				

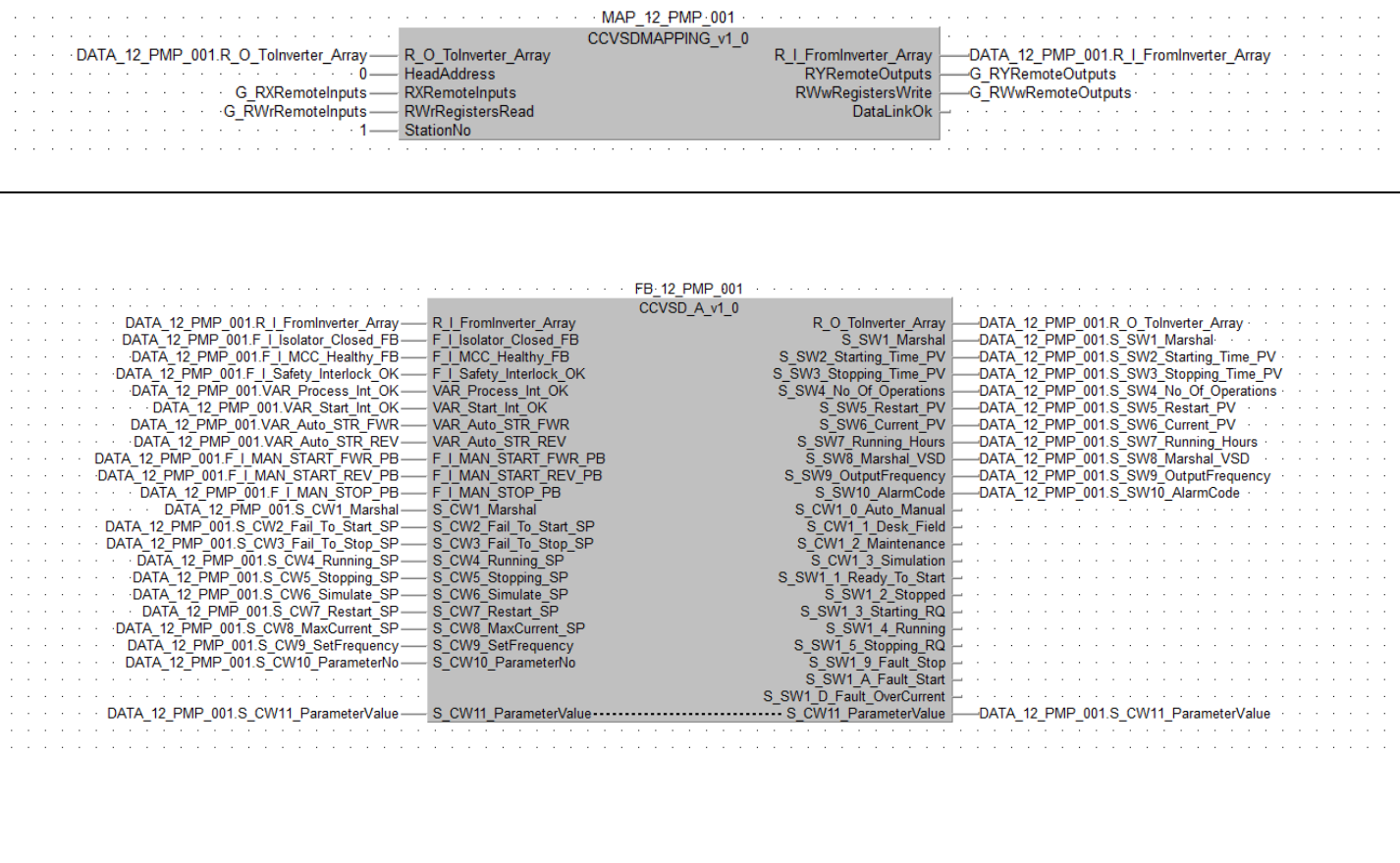
SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Process Interlock		x	x	x
Bit 1	Ready To Start		x	x	x
Bit 2	Stopped		x	x	x
Bit 3	Starting		x	x	x
Bit 4	Running		x	x	x
Bit 5	Stopping		x	x	x
Bit 6	Fault		x	x	x
Bit 7	Start Interlock OK		x	x	x
Bit 8	Isolator Closed Fault	x	x		
Bit 9	Fault Fail To Stop	x	x	x	
Bit 10	Fault Fail To Start	x	x	x	x
Bit 11	MCC Fault	x	x	x	x
Bit 12	Safety Fault	x	x	x	x
Bit 13	Overcurrent Fault	x	x	x	x
Bit 14	Reset Flash on Fault		x	x	x
Bit 15	Fail To Start due to Interlock Alarm	x	x		

SCADA Status Word1 (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	VSD Running Feedback		x	x	x
Bit 1	VSD Alarm Active	x	x	x	x
Bit 2	VSD Command Error	x	x	x	x
Bit 3	VSD Network Mode		x	x	x
Bit 4	VSD Busy		x	x	x
Bit 5	VSD Ready		x	x	x
Bit 6	VSD Overload Fault	x	x	x	x
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

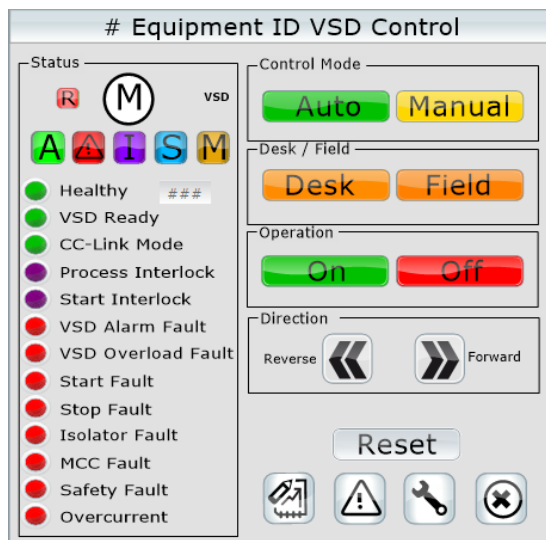
Resources	Advanced	Standard	Basic
Digital Inputs	6	5	2
Digital Outputs	0	0	0
Analog Inputs	0	0	0
Analog Outputs	0	0	0
PLC Memory Steps	871	823	747
System Words Used	84	78	65
System Bits Used	204	198	187
SCADA SCAN Points	21	17	11

Template: Advance CC-Link VSD PLC Function Block

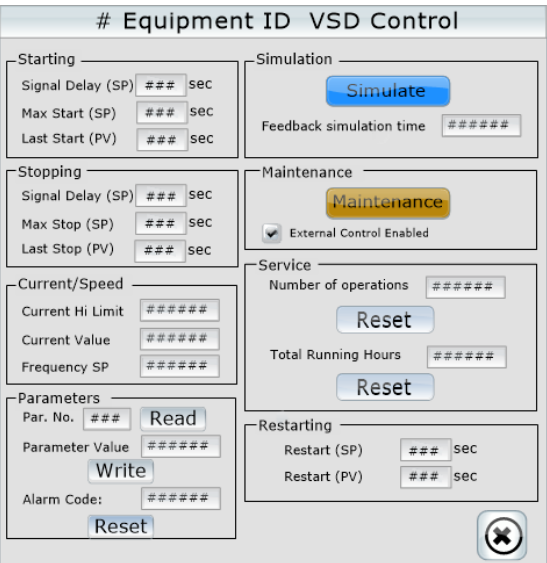
For every CC-Link VSD template used a CC-Link Mapping function block must be defined manually to map the Remote Inputs and Outputs for the CC-Link Template

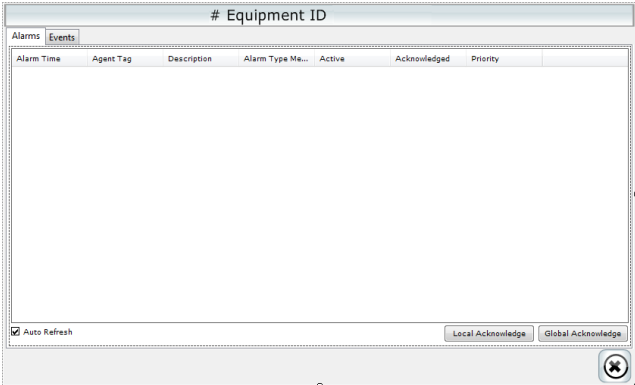


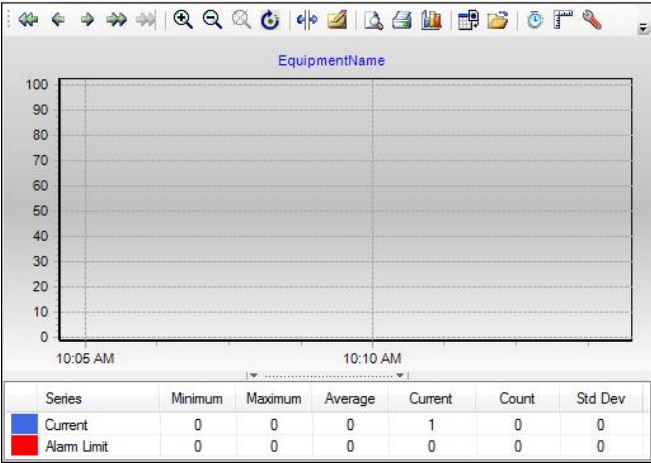
Template Graphic: Advanced Home Screen



Control	Descriptions
Auto	Auto control by PLC
Manual	Manual control by operator, desk and field mode
Desk	Manual control from the SCADA faceplate
Field	Manual control from the field start/stop station
On	Manual start the VSD
Off	Manual stop the VSD
Reset	Reset the VSD after a fault has been repaired
Healthy ###	Healthy to start, ### - Restart delay countdown in seconds
Process Interlock	VSD process ok to run interlock ready
Start Interlock	VSD ok to start interlock ready
Start Fault	VSD failed to start fault
Stop Fault	VSD failed to stop fault
Isolator Fault	VSD isolator off fault
MCC Fault	VSD MCC tripped fault
Safety Fault	VSD Safety off fault
Overcurrent	VSD overcurrent fault

Template Graphic: Advanced Setup Screen	Control	Descriptions
	Simulate	Enable simulation of VSD, disables the physical output
	Feedback Time	Time for simulation to simulate run feedback signal
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of operations	Total number of VSD start operations
	Running Hours	Total number of VSD running hours
	Reset	Reset the total counter of the current service
	Start Signal Delay (SP)	Delay the PLC to accept the VSD is running
	Max Start (SP)	Time for running signal is on after start signal is on
	Last Start (PV)	Last time for running signal was on after start signal was on
	Stop Signal Delay (SP)	Delay the PLC to accept the VSD is stopped
	Max Stop (SP)	Time running signal is off after start signal is off
	Last Stop (PV)	Last time running signal was off after start signal was off
	Current Hi Limit	Sets the value of the maximum current for the VSD to trip at
	Current Value	Actual current the VSD is running at
	Frequency SP	Frequency setpoint for VSD
	Par. No	Number of parameter to be modified.
	Parameter Value	Value of parameter that is modified
	Alarm Code	VSD alarm fault code. (see manual for descriptions)
	Restart (SP)	Restart prevention time of the VSD setpoint
	Restart (PV)	Remaining time before a VSD restart is allowed

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Current	Display the current of the VSD in the trend, with minimum, maximum and average values.
	Alarm Limit	Display the current Setpoint of alarming on high current of the VSD in the trend, with minimum, maximum and average values.

CC-Link Configuration

First you need to know which address the CC-Link card is on the PLC and the version type of the CC-Link mode. You also need to know what address ranges are available for Remote Input (X), Remote Output (Y), Read Registers (RWr) and Write Registers (RWw).

Below is an example.

My card is the 3rd module on the PLC with address 0x0030. Below it is indicated in blue.

My CC-Link Module uses the CC-Link V2 mode. This is in green.

You also need to specify the amount of Inverters connected. Below in the example we have 3. This is indicated between the green and red blocks.

My address that are available to me is: X500, Y600, D0 and D1000. See in red block

Start I/O No.	0030
Operation Setting	Operation Setting
Type	Master Station
Station No.	0
Master Station Data Link Type	PLC Parameter Auto Start
Mode	Remote Net(Ver.2 Mode)
Transmission Speed	156Kbps
Total Module Connected	3
Remote input(RX)	X500
Remote output(RY)	Y600
Remote register(RWr)	D0
Remote register(RWw)	D1000
Ver.2 Remote input(RX)	
Ver.2 Remote output(RY)	
Ver.2 Remote register(RWr)	
Ver.2 Remote register(RWw)	
Special relay(SB)	
Special register(SW)	
Retry Count	3
Automatic Reconnection Station Count	1
Standby Master Station No.	
PLC Down Select	Stop
Scan Mode Setting	Asynchronous
Delay Time Setting	0
Station Information Setting	Station Information
Remote Device Station Initial Setting	Initial Setting
Interrupt Settings	Interrupt Settings

Now you need to configure the station information of the 3 modules as was specified above.

CC-Link Station Information Module 1

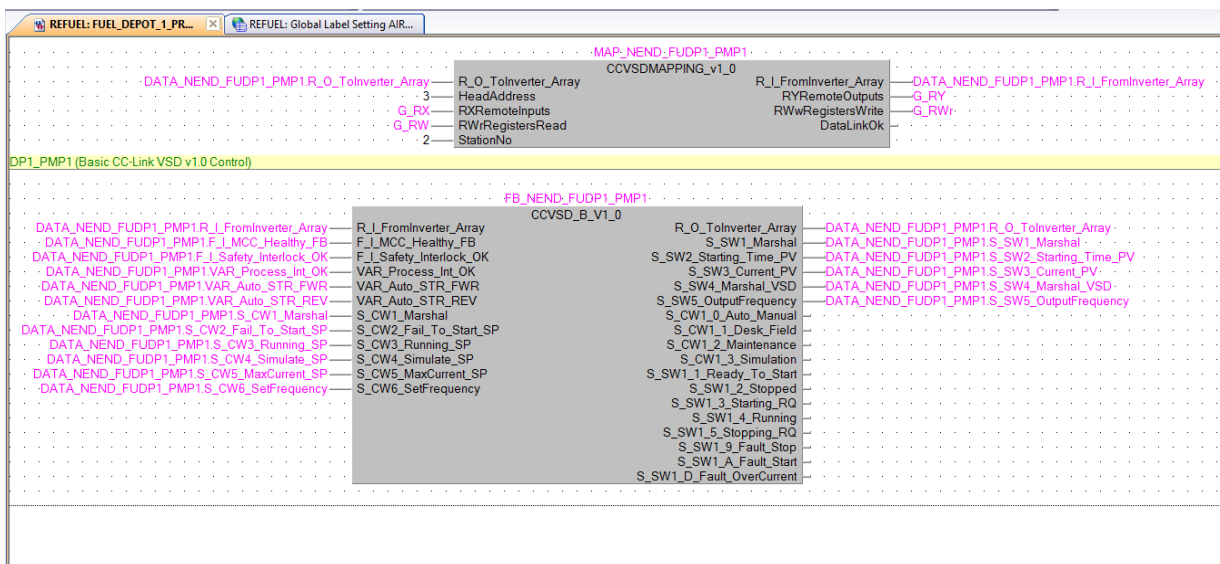
Station No.	Station Type	Expanded Cyclic Setting	Number of Occupied Stations	Remote Station Points	Reserve/Invalid Station Select	Intelligent Buffer Select(Word)		
						Send	Receive	Automatic
1/ 1	Ver.1 Remote Device Station	Single	Occupied Station 1	32Points	No Setting			
2/ 2	Ver.1 Remote Device Station	Single	Occupied Station 1	32Points	No Setting			
3/ 3	Ver.1 Remote Device Station	Single	Occupied Station 1	32Points	No Setting			

Intelligent device station at station type also includes local station and standby master station.

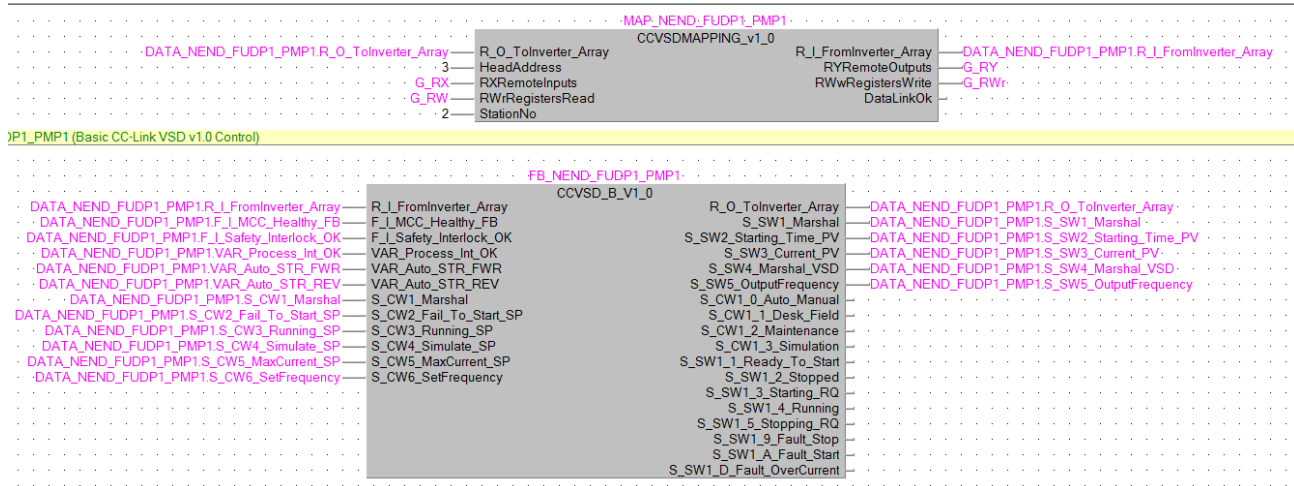
The above configuration is done because we had 3 drives connected to the PLC.

In MAPS we have a standard CC-Link FB. Now you need to setup the mapping of the CC-Link in this instance of the FB.

In order to do this you need to insert a new line above the VSD FB instance created by MAPS. Then from the MAPS FB Library go and select the CCVSDMAPPING_V1_0 and drag this into the new line.



You now need to configure the MAPPING of the CC-Link. The configuration of the MAPPING FB:



R_O_TolInverter_Array and **R_I_FromInverter_Array** – Is the signals from the DUT instance created by MAPS.

HeadAddress – is the module address of the card divide by 10. In our case it is 0x0030 / 10 which gives us 3.

RXRemoteInputs – Is it specified as G_RX. This is the X500 value we specified earlier, but the FB only accepts Words, so this is why assigned a Global label to it. The labels will be under your Global Label tree.

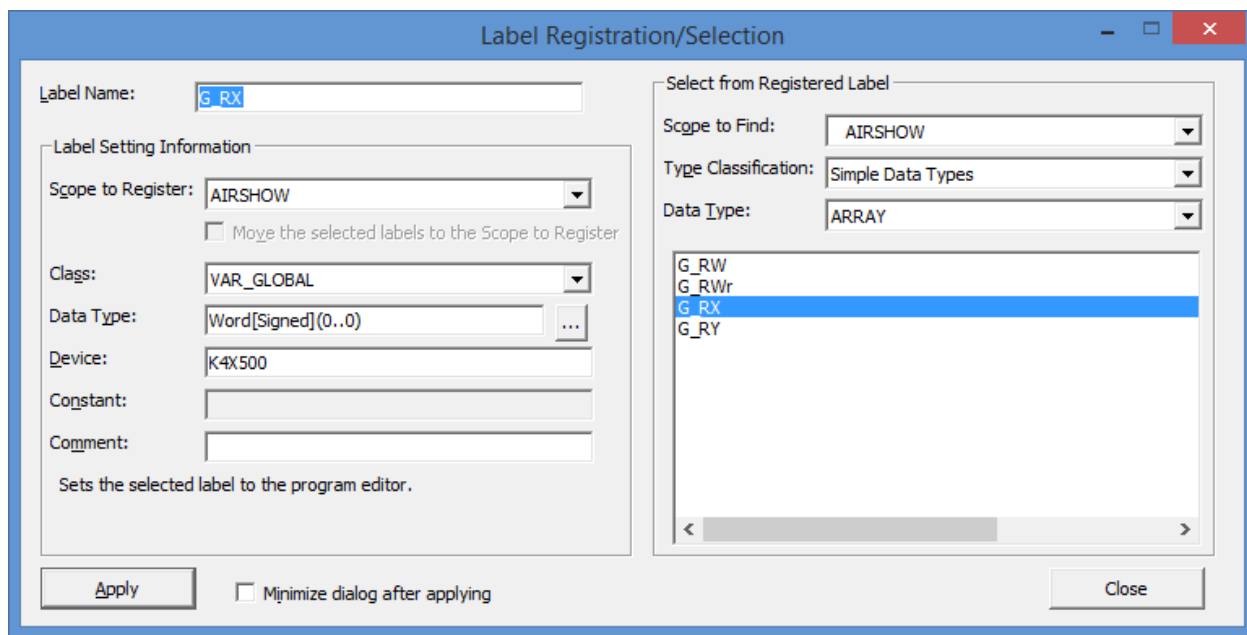
REFUEL Global Label Setting

Label Name	Data Type	Constant	Device	Address	Comment	Remark
VAR GLOBAL	DATA_NEND_FUDP1_PMP1	CCVSD_B_V1_0			NEND_FUDP1_PMP1 (Basic CC-Link VSD v1.0)	Fuel Depot 1 Pump 1
VAR GLOBAL	G_RX	Word(Signed) 0	K4X500	LDW 18 & 1200		
VAR GLOBAL	G_RY	Word(Signed) 0	K4Y500	LDW 18 & 1200		
VAR GLOBAL	G_RW	Word(Signed) 0	D0	LDW 0		
VAR GLOBAL	G_RW	Word(Signed) 0	D1000	LDW 1000		

Output

No.	Result	Data Name	Class	Content	Error Code
1	Warning	CCPR_A7NC_Advanced_0b...	Program compile	Values depending on DHO are being used as input. Results are not defined if DHO is FALSE (adder Block No. 25)	C004
2	Information	-	-	Word device (VAR range) 99 points used (Range 01289 - 01287)	F1301
3	Information	-	-	Bit device (VAR range) 415 points used (Range H7777 - H819)	F1305
4	Information	-	-	Printer (VAR range) 3 points used (Range T2046 - F2052)	F1309
5	Information	-	-	Timer (VAR range) 3 points used (Range T2045 - T2047)	F1311
6	Information	-	-	Counter (VAR range) 0 point used	F1324

Opening the label G_RX it will look like this:



Here you see that the type of the label is a Word and the device (which is value) is K4X500 which is basically X500.

Now this is exactly the same explanation for G_RW (**RW**RegistersRead), G_RY (**RY**RemoteOutputs) and G_RWr (**RW**RegistersWrite). The only remaining parameters in the FB is the StationNo.

Now the **StationNo** is the CC-Link address as set on the Inverter. Although we have 3 Modules connected we want to connect to the address 2.

You now need to compile and download the parameters to the PLC and restart the PLC.

Troubleshooting

Error Number	Error Code	Error Description
16	E.OC1	Overcurrent trip during acceleration
17	E.OC2	Overcurrent trip during constant speed
18	E.OC3	Overcurrent trip during deceleration or stop
32	E.OV1	Regenerative overvoltage trip during acceleration
33	E.OV2	Regenerative overvoltage trip during constant speed
34	E.OV3	Regenerative overvoltage trip during deceleration or stop
48	E.THT	Inverter overload trip (electronic thermal relay function)
49	E.THM	Motor overload trip(electronic thermal relay function)
64	E.FIN	Fin overheat
81	UV	Under voltage
82	E.ILF .	Input phase loss
96	E.OLT	Stall prevention
112	E. BE	Brake transistor alarm detection
128	E.GF	Output side earth(ground) fault overcurrent at start
129	E.LF	Output phase loss
144	E.OHT	External thermal relay operation
160	E.OP1	Communication option fault
176	E.PE	Parameter storage device fault
177	E.PUE	PU disconnection
178	E.RET	Retry count excess
179	E.PE2 .	Internal board fault
192	E.CPU	CPU fault
197	E.IOH .	Inrush current limit circuit fault
199	E.AIE .	Analogue input fault
235	E.13	Internal circuit fault
241	E. 1	Option fault
246	E. 6	CPU fault
247	E. 7	CPU fault